

Tribhuvan University
Faculty of Humanities and Social Sciences
Koshi Saint James College, Itahari
Mid Term Exam – 2081

Bachelor in Computer Application

Course Title: Title: Object Oriented Programming in Java

Code No: CACS 204

Semester 3rd

Full Marks: 60

Pass Marks: 24

Time 3:00 hours

Candidates are required to answer the questions in their own words as far as possible.

B. Attempt any question.

[5x6=30]

2. What is Java? Explain the principle of OOP with Java.
3. Explain the difference between method overloading and method overriding in Java.
Provide examples to illustrate each concept.
4. What is data type? Explain fundamental data types with suitable example.
5. Explain the I/O package.
6. Write a factorial program to demonstrate the concept of method recursion.
7. What are the different types of classes available in java
8. What is Object? How it is created in java? Explain.
9. What is Constructor? Describe their types..

Attempt any two questions.

[10x2=20]

11. What is exception handling? Explain all the keywords of Exception handling with a demonstration.
12. Write a program in java display the multiplication table of a given integer.
13. Define inheritance. Discuss benefits of using inheritance. Discuss multiple inheritance with suitable example.

The End

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Bachelor in Computer Application
Course Title: Title: Web Technology
Code No: CACS 205
Semester 3rd

Full Marks: 60
Pass Marks: 24
Time 3:00 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6 x 5 =30]

1. What is CSS? Discuss types of CSS with example.
2. What is HTML element & attribute, describe global attribute of HTML?
3. Describe image tag with different attribute?
4. Explain sup and sub tag in HTML with suitable example?
5. What is XML? Explain the elements and attributes of XML.
6. What is Regex? Write Regex command that accept first capital letter and last letter must be '#'.
#
7. Write HTML tag to generate the following output.

- FOHSS
 - BCA
 - 1. IoT
- FOM
 - 1. BIM
 - 2. IoT

8. Write HTML Tag to generate the following table.

Fill the form below	
Name	
Password	
Feedback	
Gender	☐ Male ☐ Female
Subject	☐ Web ☐ Math ☐ Graphics ☐ English
	Reset All Submit Above Details

Group C

Attempt any TWO questions.

[2 x 10=20]

9. What is CSS selectors? Describe different types of CSS selector with example.
10. What is Tier Architecture? Explain Two Tier and Multi-Tier Architecture with its block diagram.
11. Explain XML Namespaces and Schema with suitable example.

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Bachelor in Computer Application

Course Title: System Analysis and Design

Code No: CACS 202

Semester 3rd

Full Marks: 60

Pass Marks: 24

Time 3:00 hours

Candidates are required to answer the questions in their own words as far as possible.

Group 'B'

Attempt any SIX questions:

[5x6=30]

2. Write and describe the different types of project feasibility factors.
3. Define case tools with its advantages and disadvantages
4. Who is system analyst? Discuss the roles of system analyst.
5. Explain the types of Information system.
6. What are the advantages and disadvantages of discovery prototyping?
7. What is system requirement analysis? Write
8. Describe the project identification and selection process.
9. Describe four traditional techniques for collecting information during analysis with advantages.

Group 'C'

Attempt any TWO questions:

[2 x 10 = 20]

9. What is JAD? How is it better than traditional information gathering techniques?
What are its weakness?
10. Explain different phases of SDLC. Explain each of them in detail.
11. Define DFD. Write a rules to draw a DFD. Draw a DFD upto level 1 for library management system.

The End

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Mid Term Exam – 2081

Bachelor in Computer Application
Course Title: Data Structure & Algorithm
Code No: CACS 201
Semester 3rd

Full Marks: 60
Pass Marks: 24
Time 3:00 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5=30]

2. What do you mean by ADT? Explain space and time complexity of the algorithm in detail. [1+4]
3. How can you convert given infix expression to their equivalent prefix expression? [5]
4. How circular queue differ from the linear queue. Explain. [5]
5. Mention how to insert and delete the first node from the singly linked list. [5]
6. Define a linked list. Explain the array implementation of linked lists. [1+4]
7. What do you mean by selection sort? How does selection sort work? Explain with suitable example. [1+4]
8. Illustrate the stack as a linked list and the queue as a linked list. [2.5+2.5=5]

Group C

Attempt any TWO questions:

[10×2=20]

9. Show the stack as ADT. Mention the basic operation of the stack. And also describe push and pop function of stack along with its code implementation. [4+6]
10. Define data structure. Mention the importance of data structure. Also write the operations performed on the data structure. Explain the types of data structure in brief. [1+1+3+5]
11. Define sorting. Sort the following data items by using a) heap sort b) Quick sort c) Merge sort. Data items are: [4,5,6,88,1,22,43,56,53,2,11,57,9] [1+3+3+3]

The End

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Mid Term Exam – 2081

Bachelor in Computer Application
Course Title: Probability and statistics
Code No: CAST 202
Semester 3rd

Full Marks: 60
Pass Marks: 24
Time 3:00 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any six questions

[6x5=30]

- ✓11. Write down the process of collecting primary data
✓12. A frequency distribution of marks of 100 students is given below. Frequencies corresponding to two groups are missing from the table. The median is known to be 49.5 marks.

Marks	0-19	20-39	40-59	60-79	80-99
No. of students	14	?	26	?	16

- i. Find the missing frequencies.
ii. Calculate the limits of marks obtained by middle 60% students.
13. The mean and variance of the marks in statistics obtained by all the 50 students of a certain college was computed as 60 and 100 respectively. Later on it was discovered that the scored 76 was wrongly taken as 67. Find the correct mean and standard deviation. Also calculate the coefficient of variation of marks after ignoring the wrong value?
14. Define correlation. Calculate and analyze the correlation coefficient between the Price of A and B

Price of A	25	28	35	20	22	30	31	22
Price of B	35	39	48	29	30	38	40	32

Also find the coefficient of determination and interpret it.

- ✓15. The following table gives information on ages in cholesterol levels for a random sample of 10 men.

Age (Years)	58	69	43	39	63	52	47	31	74	36
Cholesterol Level in suitable units	189	235	193	177	154	191	213	165	198	181

- a) Find a regression equation of cholesterol level on age.
b) Estimate the value of cholesterol level when age is 60 years.

- ✓16. In a bag, there are 5 red balls and 4 green balls. Two balls are drawn at random.
i. Find the probability that both balls are green.
ii. Find the probability that both are of different colors.
iii. Find the probability that both balls are red if the ball is drawn one after another with replacement.

17. Calculate the Rank Correlation from the following table and interpret the result obtained.

Marks in Maths	15	20	28	12	40	60	20	80
Marks in Science	40	30	50	30	20	10	30	60

Group C

Attempt any two questions

[2x10=20]

18. An analysis of monthly wages paid to worker in two firms A and B belonging to the same Industry gives the following results

	Firm A	Firm B
No. of wage earners	550	650
Average monthly wages	50	45
Standard deviation of the distribution of wage	$\sqrt{90}$	$\sqrt{120}$

- Which firm A or B pays a larger amount as monthly wages?
- In which firm A or B has there greater variability in individual wages?
- Calculate the mean and standard derivation of wages of all workers of the firm A and B taken together

19. While calculating the correlation coefficient between two variables X and Y from 25 pair of observations, the following results were obtained.

$$N=25, \sum X = 125, \sum X^2 = 650, \sum Y = 100, \sum Y^2 = 460 \text{ and } \sum XY = 508.$$

At the time of final rechecking the calculated data, the pairs (x= 8, 6 and Y= 12, 8) were misread as (X= 18, 16 and Y=14, 16). Now, obtain the correct correlation coefficient by correcting the values. Also, calculate the limits of population.

20. From the following bi-variate frequency table, find out if there exists any relationship between advertisement expenditure (in 00 Rs.) and sales revenue (in 000 Rs.) and test the significance of the result. Also estimate sales revenue when advertisement expenditure is Rs. 40,000.

Expenditure (in 00 Rs.)	Sales Revenue (in 000 Rs.)				
	0-50	50-100	100-150	150-200	200-250
0-40	12	6	8	-	-
40-80	2	18	4	5	1
80-120	-	8	10	2	4
120-160	-	1	10	2	1
160-200	-	-	1	2	3

Best of Luck